

Capstone Project Proposal

EDGAR Intelligence Platform

SEC Filing Search, Analytics, and AI Research Assistant

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Problem and Target Users

SEC EDGAR is the authoritative source for U.S. public company filings, but it is hard to actually use. The data is scattered across raw HTML documents, XBRL files, and archive directories, with no easy way to search across companies, compare financial metrics over time, or ask plain-language questions about a filing. Analysts end up opening filings one at a time and copying numbers into spreadsheets.

Target users: equity research analysts who compare financials across companies and quarters; retail investors doing fundamental research; and compliance or corporate-research teams who monitor filing activity.

Primary workflow (end-to-end user journey): a user searches for a company, browses its filing history, opens a filing to view extracted financials and sections, compares metrics across periods or peers on a dashboard, then asks the AI agent to summarize the filing and save findings to a watchlist or research note. The platform turns EDGAR from a document repository into a searchable financial data platform. It is a mini financial data platform, not a web scraper.

As a worked example, the platform will ingest and process a filing such as Alphabet's Q2 2026 10-Q. That single filing contains the primary HTML document, exhibits, XBRL instance data, taxonomy files, and a full submission text file, showing the mix of structured and unstructured data the project handles.

Data Sources and Integration Details

The project uses SEC EDGAR as the primary source and a market-data API for enrichment. Ingestion uses the SEC's official APIs and archives rather than emulating a browser, and respects the SEC's automated-access rules (10 requests per second, declared User-Agent header).

Source / Endpoint	Auth	Rate limit	Format	Data	Update freq.
EDGAR Submissions API (data.sec.gov/submissions/CIK...json)	None; User-Agent	10 req/s	JSON	Filing history + metadata	Daily / on demand
XBRL company facts (data.sec.gov/api/xbrl/companyfacts/...)	None; User-Agent	10 req/s	JSON	Extracted financial facts	Daily
EDGAR archives (sec.gov/Archives/edgar/data/...)	None; User-Agent	10 req/s	HTML/ XML/XBRL/ txt	Full filing documents	Per filing
Market Data API (Alpha Vantage / Finnhub / yfinance)	API key	5 to 60 req/min (free)	JSON	Price, volume, mkt cap, history	Daily bars

Ingestion into Bronze: a rate-limited PySpark job pulls submissions and XBRL facts per CIK, downloads referenced filing documents from the archives, and writes each stream to raw Bronze Delta tables. Original

documents are retained so SEC data can be reproduced and reprocessed. The market-data API is polled daily and landed alongside the SEC data for enrichment in Silver.

Spark Data Pipeline

Built with Apache Spark and PySpark on Databricks using a medallion architecture.

Bronze (raw)

Raw EDGAR and market data retained verbatim: bronze_company_submissions, bronze_filings, bronze_filing_documents, bronze_xbrl_facts, bronze_filing_text.

Silver (cleaned and normalized)

Key transformations: parse XBRL instance and taxonomy files into typed facts, parse filing HTML into sections, entity resolution to standardize company identifiers (CIK to ticker to name), deduplicate filings by accession number, normalize financial concepts and reporting periods, extract financial tables and values, and chunk filing text for search. Tables: silver_companies, silver_filings, silver_filing_sections, silver_financial_facts, silver_financial_periods, silver_exhibits.

Gold (application-ready)

Analytical datasets used by the app and analytics layer: gold_company_financials, gold_revenue_history, gold_income_statement_metrics, gold_balance_sheet_metrics, gold_cash_flow_metrics, gold_filing_activity, gold_company_comparisons.

Partitioning and quality checks

Fact and filing tables are partitioned by filing period (year, quarter) and clustered by CIK for fast company-level reads. Quality checks: not-null constraints on keys (CIK, accession number, period), deduplication on accession number, range checks on numeric facts, and referential checks that every fact resolves to a known company and filing before promotion to Gold.

Lakebase Data Model

Lakebase is the operational relational database (managed PostgreSQL) for the app, giving the project a true transactional component instead of using Delta for everything. Core tables: users, companies, watchlists, watchlist_companies, saved_filings, saved_research, agent_conversations, agent_actions.

watchlists

Column	Description
watchlist_id	Unique watchlist identifier
user_id	Application user
name	Watchlist name
created_at	Creation timestamp

watchlist_companies

Column	Description
watchlist_id	Parent watchlist

Column	Description
cik	SEC CIK
ticker	Stock ticker
added_at	Timestamp added

saved_research

Column	Description
research_id	Research record identifier
user_id	User
company_cik	Related company
filing_id	Related filing
title	Research title
notes	User notes
created_at	Timestamp

Action-Taking AI Agent

The app includes an SEC Research Assistant, a tool-calling agent that both retrieves information and performs write actions. Retrieval reads from Gold Delta and Lakebase; writes go to Lakebase.

Retrieval tools

- search_company()
- search_filings()
- get_filing()
- get_financial_metric()
- compare_companies()
- search_filing_text()
- get_saved_research()

Write / action tools

- save_filing()
- save_company_to_watchlist()
- create_research_note()
- update_research_note()
- remove_from_watchlist()

Example interaction

User: "Find Alphabet's most recent 10-Q and summarize the major changes in revenue and operating expenses." The agent retrieves the filing and financial data and answers.

User: "Save this filing to my research list and add a note that I want to compare Google's cloud growth against Microsoft." The agent writes those records to Lakebase.

Analytics Pipeline

Changes to the Lakebase operational tables are captured with Lakebase Change Data Feed and propagated into Delta analytics tables. If CDF from Lakebase is unavailable in the target workspace, the same tables are populated with Delta Live Tables, which the requirements allow as an alternative.

Change events captured: inserts and updates to watchlists, saved filings, research notes, agent conversations, and agent actions.

Transformation: change rows are appended to append-only event tables, then aggregated into usage and activity marts.

Metrics and dashboards:

- Most searched companies, most viewed filings, most requested filing types
 - Most viewed financial metrics
 - Saved filings and watchlist activity over time
 - AI agent usage, per-tool call counts, and agent action success rate
 - Query latency and usage funnels (search to filing view to save)
 - Research notes created and filings searched per user
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Frontend and Deployment

Deployed as a working Databricks App. Databricks Apps can use Lakebase directly as a managed PostgreSQL backend, which fits this architecture.

- Company Search: search by name or ticker (Alphabet, Microsoft, Apple, Amazon, Tesla); view company info, recent filings, filing history, headline metrics. Reads Gold Delta and Lakebase.
- Filing Explorer: open a filing (for example Alphabet 10-Q, Q2 2026); view metadata, sections, financial statements, XBRL facts, exhibits, filing text, and a link to the SEC source.
- Financial Dashboard: visualize revenue, net income, operating income, cash flow, assets, liabilities, debt, EPS, and year-over-year changes. Reads Gold Delta.
- AI Research Assistant: chat panel calling the tool-calling agent; the agent reads Gold Delta and Lakebase and writes user actions to Lakebase.

Deployment plan: package the app with its app.yaml and dependencies, register it as a Databricks App, bind it to the Lakebase instance and Unity Catalog resources, and use the workspace identity for auth so the app queries Gold Delta and Lakebase under governed permissions.

Big Data Characteristics (meets Variety + Volume)

The project meets two of the three Vs: Variety and Volume.

Variety (qualifies)

SEC filings arrive in JSON, HTML, XML, XBRL, plain text, financial tables, exhibits, and narrative sections. A single Alphabet 10-Q includes an HTML filing, full submission text, XBRL instance data, and XBRL schema, calculation, definition, label, and presentation files, plus multiple exhibits. The pipeline processes both structured and unstructured or semi-structured data.

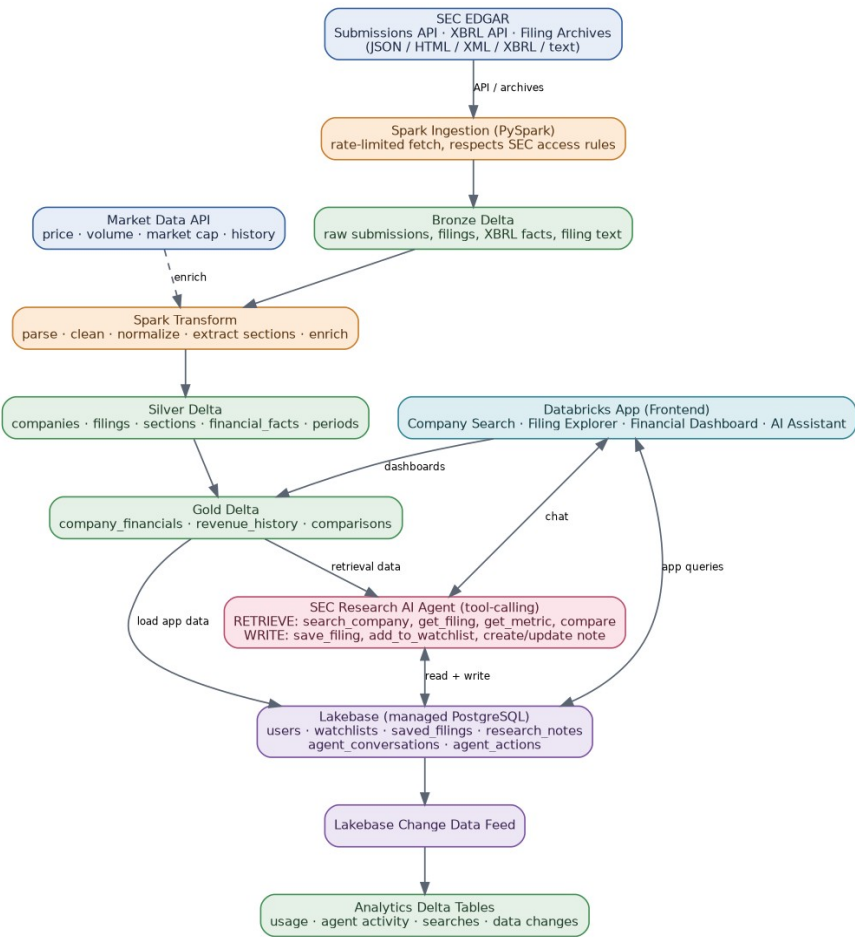
Volume (qualifies, over 1M rows)

The row count is anchored in XBRL financial facts. Scoping to roughly 500 companies (S&P 500) across 10 or more years of 10-K, 10-Q, and 8-K filings gives about 500 companies times about 40 filing periods times several hundred to a few thousand XBRL facts per filing. At a conservative 800 facts per filing, that is roughly 16 million rows in silver_financial_facts alone. Two more tables independently exceed one million rows:

- Filing-text chunks: hundreds of searchable chunks per filing across tens of thousands of filings.
- Daily market-data bars: about 500 companies times about 2,500 trading days is roughly 1.25 million rows.

Even a reduced pilot of 100 companies clears one million XBRL facts, so the Volume claim is credible at full scope and at pilot scope.

Proposed Architecture



Technology Stack

Component	Technology
Data Source	SEC EDGAR (Submissions API, XBRL API, filing archives)

Component	Technology
External API	Market Data API (Alpha Vantage, Finnhub, or yfinance)
Processing	Apache Spark / PySpark
Lakehouse	Delta Lake (Bronze / Silver / Gold)
Operational DB	Lakebase (managed PostgreSQL)
Change Data Capture	Lakebase Change Data Feed (DLT as fallback)
AI Agent	Databricks tool-calling agent
Frontend / Deployment	Databricks Apps
Analytics	Databricks SQL / Delta
Governance	Unity Catalog
Language	Python

Expected Outcome

The final application provides a single interface for researching SEC filings without manually navigating EDGAR. A user can search for a company, view its historical filings, open and explore individual filings and extracted financials, compare metrics across periods or companies, ask an AI agent questions, save filings and research notes through the agent, create watchlists, and view analytics about their own research activity.

The project demonstrates an end-to-end data engineering and AI architecture using SEC EDGAR, Apache Spark, Delta Lake, Lakebase, Change Data Feed, an external market API, a tool-calling AI agent, and a deployed Databricks App.